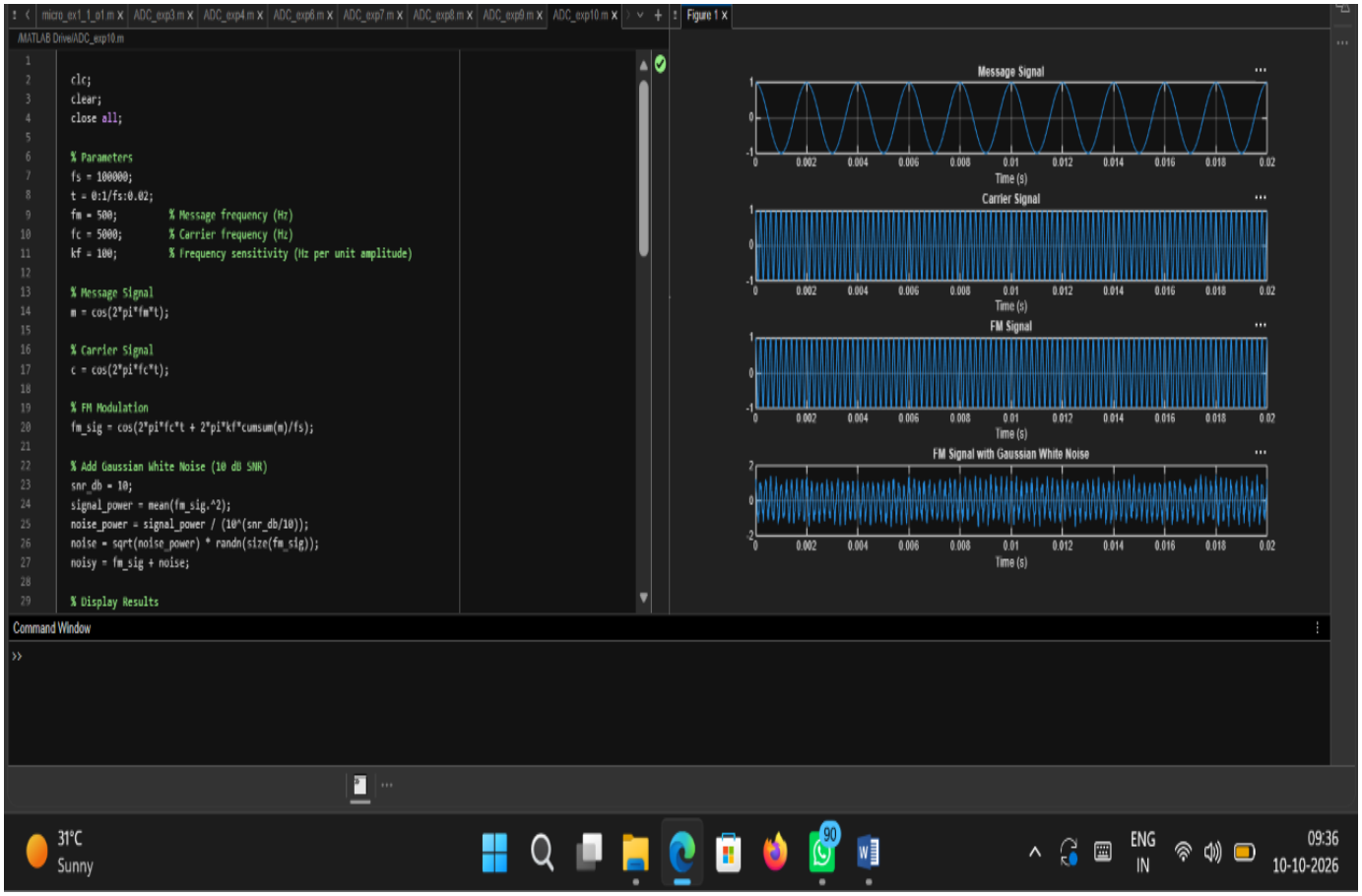
Ex.No:10: PERFORMANCE ANALYSIS OF FM SYSTEM UNDER GAUSSIAN WHITE NOISE USING MATLAB

Objective 1:



Code:

```
clc;

clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;           % Message frequency (Hz)
fc = 5000;          % Carrier frequency (Hz)
kf = 100;           % Frequency sensitivity (Hz per unit amplitude)

% Message Signal
m = cos(2*pi*fm*t);

% Carrier Signal
c = cos(2*pi*fc*t);

% FM Modulation
```

```
fm_sig = cos(2*pi*fc*t + 2*pi*kf*cumsum(m)/fs);

% Add Gaussian White Noise (10 dB SNR)
snr_db = 10;
signal_power = mean(fm_sig.^2);
noise_power = signal_power / (10^(snr_db/10));
noise = sqrt(noise_power) * randn(size(fm_sig));
noisy = fm_sig + noise;

% Display Results
figure;

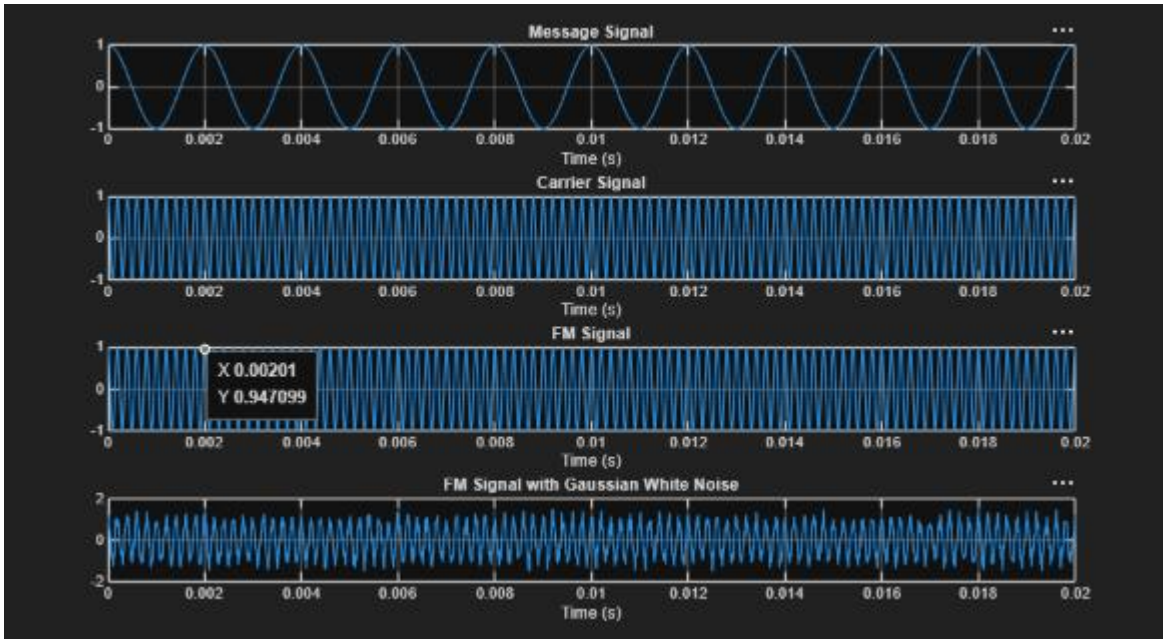
subplot(4,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
grid on;

subplot(4,1,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
grid on;

subplot(4,1,3);
plot(t,fm_sig);
title('FM Signal');
xlabel('Time (s)');
grid on;

subplot(4,1,4);
plot(t,noisy);
title('FM Signal with Gaussian White Noise');
xlabel('Time (s)');
grid on;
```

Output:


Message Signal
Time (s)
Carrier Signal
Time (s)
FM Signal
X 0.00201
Y 0.947099
Time (s)
FM Signal with Gaussian White Noise
Time (s)